7 **(a)** The emission line spectrum consists of a series of distinct lines of specific colours/ wavelengths/ frequencies on a dark background.

Each line is produced by the emission of photons of a specific quantum of energy. Since the photons have specific energy, they must be emitted when electrons in the atoms de-excites between discrete energy levels.

(b) **(i)** The characteristic K_α and K_β lines are due to X-ray photons produced from electron transitions from shell L to K and shell M to K, respectively (or from energy level n = 2 to n = 1 and n = 3 to n = 1, respectively).

Since shell L is nearer to shell K than shell M to shell K, K_α photons have smaller energy E than K_β photons. As energy E is inversely proportional to the wavelength λ, K_α photons have a longer wavelength than K_β photons.

(ii) X-ray photons are emitted when high speed electrons collide with the target atom and undergo many rapid decelerations. The kinetic energy lost by an incident electron is converted into the energy of an X-ray photon. The amount of energy lost varies since an electron can undergo multiple collisions resulting in X-ray photons of varying energies.

The most energetic X-ray photon is produced when an incident electron loses all its kinetic energy in one collision. Since the energy of a photon is given by $E = \dfrac{hc}{\lambda}$, this most energetic X-ray photon has the shortest wavelength.

Raffles Institution
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8 **(c)** **(i)**